THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Surface Areas and Volumes** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 858 cm²	TSA = CSA of cone + CSA of hemisphere $= \pi rl + 2\pi r^2$; TSA $= \frac{22}{7} \times 7 \times 25 + 2 \times \frac{22}{7} \times 7^2 = 858$ cm²
2	**(d)** 374 cm²	$2\pi rh + 2(2\pi r^2) = 2 \times \frac{22}{7} \times 3.5 \times 10 + 4 \times \frac{22}{7} \times 3.5^2$; TSA = 374 cm²
3	**(c)** 1330 cm²	$6a^2 - \pi r^2 + 2\pi r^2 = 6a^2 + \pi r^2$ with $r = 7$ cm; TSA $= 1176 + 154 = 1330$ cm²
4	**(a)** 1760 cm²	TSA = CSA cylinder + one flat base + CSA of cone; $= 1056 + 154 + 550 = 1760$ cm²
5	**(d)** 374 cm²	$2\pi rh + 2(2\pi r^2)$ with $r = 3.5$, $h = 10$; TSA $= 220 + 154 = 374$ cm²
6	**(c)** 32	Volume of a sphere $= \frac{4}{3}\pi(3)^3$; $n = \frac{3r^2h}{4\rho^3}$; $n = 32$
7	**(a)** 6 cm	$\frac{4}{3}\pi R^3 = \frac{1}{3}\pi(6)^2(24)$ gives $R^3 = 216$; $R = 6$ cm
8	**(c)** 5082 cm²	TSA = CSA of cone + CSA of hemisphere $= \pi rl + 2\pi r^2$; TSA $= \frac{22}{7} \times 21 \times 35 + 2 \times \frac{22}{7} \times 21^2 = 5082$ cm²
9	**(d)** 1232 cm²	$2\pi rh + 2(2\pi r^2) = 2 \times \frac{22}{7} \times 7 \times 14 + 4 \times \frac{22}{7} \times 7^2$; TSA = 1232 cm²
10	**(b)** 5320 cm²	$6a^2 - \pi r^2 + 2\pi r^2 = 6a^2 + \pi r^2$ with $r = 14$ cm; TSA $= 4704 + 616 = 5320$ cm²

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Volume of sphere = volume of wire (cylinder) $= \pi r^2 L$; $\frac{4}{3}\pi(6)^3 = \pi(2)^2 L$ (1); $L = 72$ cm (1).	1+1
12	Volume of cuboid $= n\times$ volume of a cube; $n = \frac{24\times12\times9}{6^3}$ (1); $n = 12$ (1).	1+1
13	Volume of water displaced = volume of sphere = rise × base area; $\frac{4}{3}\pi(3)^3 = \pi(3)^2 x$ (1); $x = 4$ cm (1).	1+1
14	Volume of sphere = volume of wire (cylinder) $= \pi r^2 L$; $\frac{4}{3}\pi(6)^3 = \pi(3)^2 L$ (1); $L = 32$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Volume = hemisphere + cone (1); Hemisphere $= \frac{2}{3}\pi r^3 = 19404$ cm³; cone $= \frac{1}{3}\pi r^2 h = 4620$ cm³ (1); Volume = 24024 cm³ (1).	1+1+1
16	Volume = cylinder + two hemispheres $= \pi r^2 h + \frac{4}{3}\pi r^3$ (1); Cylinder = 3465 cm³; two hemispheres = 4851 cm³ (1); Volume = 8316 cm³ (1).	1+1+1
17	Volume left = cylinder − cone (1); Cylinder = 1848 cm³; cone = 616 cm³ (1); Volume left = 1232 cm³ (1).	1+1+1
	OR: Volume = hemisphere + cone (1); Hemisphere $= \frac{2}{3}\pi r^3 = 19404$ cm³; cone $= \frac{1}{3}\pi r^2 h = 9240$ cm³ (1); Volume = 28644 cm³ (1).	1+1+1
18	Volume = cylinder + two hemispheres $= \pi r^2 h + \frac{4}{3}\pi r^3$ (1); Cylinder = 6930 cm³; two hemispheres = 4851 cm³ (1); Volume = 11781 cm³ (1).	1+1+1
	OR: Volume left = cylinder − cone (1); Cylinder = 3696 cm³; cone = 1232 cm³ (1); Volume left = 2464 cm³ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Volume of water is the same in both tanks (2); $\pi r^2 h = 1540$ cm³ and base area of cuboid = 77 cm² (2); Height = $1540 \div 77 = 20$ cm (1).	2+2+1
	OR: Volume of water in the cone $= \frac{1}{3}\pi r^2 h$ (2); $\frac{1}{3}\pi(6)^2(8) = \pi(4)^2 x$ (2); $x = 6$ cm (1).	2+2+1
20	Volume of water is the same in both tanks (2); $\pi r^2 h = 1848$ cm³ and base area of cuboid = 308 cm² (2); Height = $1848 \div 308 = 6$ cm (1).	2+2+1
	OR: Volume of water in the cone $= \frac{1}{3}\pi r^2 h$ (2); $\frac{1}{3}\pi(6)^2(12) = \pi(3)^2 x$ (2); $x = 16$ cm (1).	2+2+1

Mistakes to avoid:

- Counting the joined faces in the total surface area of a combination of solids (e.g. adding the base of a cone resting on a hemisphere).
- Using the vertical height instead of the *slant* height in the CSA and TSA of a cone.

- Forgetting the $\frac{1}{3}$ in the volume of a cone or using $2\pi r^2$ for the surface area of a full sphere.
- Assuming surface area stays constant when a solid is melted and recast; only the volume is conserved.
- Mixing units (cm, m, mm) or writing cm instead of cm^2/cm^3 in the answer.
- Using diameter in place of radius while substituting into a formula.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Surface Areas and Volumes** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 7040 cm²	TSA = CSA cylinder + one flat base + CSA of cone; $= 4224 + 616 + 2200 = 7040$ cm^2
2	**(d)** 1276 cm²	$2\pi rh + 2(2\pi r^2)$ with $r = 7$, $h = 15$; TSA $= 660 + 616 = 1276$ cm^2
3	**(b)** 81	Volume of a sphere $= \frac{4}{3}\pi(2)^3$; $n = \frac{3r^2h}{4\rho^3}$; $n = 81$
4	**(a)** 3 cm	$\frac{4}{3}\pi R^3 = \frac{1}{3}\pi(3)^2(12)$ gives $R^3 = 27$; $R = 3$ cm
5	**(b)** 11770 cm²	TSA = CSA of cone + CSA of hemisphere $= \pi rl + 2\pi r^2$; TSA $= \frac{22}{7} \times 35 \times 37 + 2 \times \frac{22}{7} \times 35^2 = 11770$ cm^2
6	**(a)** 1496 cm²	$2\pi rh + 2(2\pi r^2) = 2 \times \frac{22}{7} \times 7 \times 20 + 4 \times \frac{22}{7} \times 7^2$; TSA $= 1496$ cm^2
7	**(c)** 332.5 cm²	$6a^2 - \pi r^2 + 2\pi r^2 = 6a^2 + \pi r^2$ with $r = 3.5$ cm; TSA $= 294 + 38.5 = 332.5$ cm^2
8	**(a)** 7392 cm²	TSA = CSA cylinder + one flat base + CSA of cone; $= 3696 + 1386 + 2310 = 7392$ cm^2
9	**(b)** 1496 cm²	$2\pi rh + 2(2\pi r^2)$ with $r = 7$, $h = 20$; TSA $= 880 + 616 = 1496$ cm^2
10	**(d)** 27	Volume of a sphere $= \frac{4}{3}\pi(2)^3$; $n = \frac{3r^2h}{4\rho^3}$; $n = 27$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Volume of cuboid $= n\times$ volume of a cube; $n = \frac{24\times12\times15}{3^3}$ (1); $n = 160$ (1).	1+1
12	Volume of water displaced = volume of sphere = rise × base area; $\frac{4}{3}\pi(6)^3 = \pi(4)^2x$ (1); $x = 18$ cm (1).	1+1
13	Volume of sphere = volume of wire (cylinder) $= \pi r^2L$; $\frac{4}{3}\pi(3)^3 = \pi(1)^2L$ (1); $L = 36$ cm (1).	1+1
14	Volume of cuboid $= n\times$ volume of a cube; $n = \frac{24\times18\times12}{6^3}$ (1); $n = 24$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Volume = hemisphere + cone (1); Hemisphere $= \frac{2}{3}\pi r^3 = 2425.5$ cm^3; cone $= \frac{1}{3}\pi r^2h = 1617$ cm^3 (1); Volume $= 4042.5$ cm^3 (1).	1+1+1
16	Volume = cylinder + two hemispheres $= \pi r^2h + \frac{4}{3}\pi r^3$ (1); Cylinder $= 20790$ cm^3; two hemispheres $= 38808$ cm^3 (1); Volume $= 59598$ cm^3 (1).	1+1+1
17	Volume left = cylinder − cone (1); Cylinder $= 2310$ cm^3; cone $= 770$ cm^3 (1); Volume left $= 1540$ cm^3 (1).	1+1+1
	OR: Volume = hemisphere + cone (1); Hemisphere $= \frac{2}{3}\pi r^3 = 2425.5$ cm^3; cone $= \frac{1}{3}\pi r^2h = 3234$ cm^3 (1); Volume $= 5659.5$ cm^3 (1).	1+1+1
18	Volume = cylinder + two hemispheres $= \pi r^2h + \frac{4}{3}\pi r^3$ (1); Cylinder $= 41580$ cm^3; two hemispheres $= 38808$ cm^3 (1); Volume $= 80388$ cm^3 (1).	1+1+1
	OR: Volume left = cylinder − cone (1); Cylinder $= 5544$ cm^3; cone $= 1848$ cm^3 (1); Volume left $= 3696$ cm^3 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Volume of water is the same in both tanks (2); $\pi r^2h = 2310$ cm^3 and base area of cuboid $= 231$ cm^2 (2); Height $= 2310 \div 231 = 10$ cm (1).	2+2+1
	OR: Volume of water in the cone $= \frac{1}{3}\pi r^2h$ (2); $\frac{1}{3}\pi(6)^2(15) = \pi(5)^2x$ (2); $x = 7.2$ cm (1).	2+2+1
20	Volume of water is the same in both tanks (2); $\pi r^2h = 3696$ cm^3 and base area of cuboid $= 462$ cm^2 (2); Height $= 3696 \div 462 = 8$ cm (1).	2+2+1
	OR: Volume of water in the cone $= \frac{1}{3}\pi r^2h$ (2); $\frac{1}{3}\pi(6)^2(10) = \pi(5)^2x$ (2); $x = 4.8$ cm (1).	2+2+1

Mistakes to avoid:

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- Using the vertical height instead of the *slant* height in the CSA and TSA of a cone.
- Forgetting the $\frac{1}{3}$ in the volume of a cone or using $2\pi r^2$ for the surface area of a full sphere.
- Assuming surface area stays constant when a solid is melted and recast; only the volume is conserved.
- Mixing units (cm, m, mm) or writing cm instead of cm^2/cm^3 in the answer.
- Using diameter in place of radius while substituting into a formula.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Surface Areas and Volumes** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 6 cm	$\frac{4}{3}\pi R^3 = \frac{1}{3}\pi(12)^2(6)$ gives $R^3 = 216$; $R = 6$ cm
2	**(b)** 3432 cm²	TSA = CSA of cone + CSA of hemisphere $= \pi rl + 2\pi r^2$; TSA $= \frac{22}{7} \times 14 \times 50 + 2 \times \frac{22}{7} \times 14^2 = 3432$ cm^2
3	**(c)** 4928 cm²	$2\pi rh + 2(2\pi r^2) = 2 \times \frac{22}{7} \times 14 \times 28 + 4 \times \frac{22}{7} \times 14^2$; TSA $= 4928$ cm^2
4	**(d)** 2992.5 cm²	$6a^2 - \pi r^2 + 2\pi r^2 = 6a^2 + \pi r^2$ with $r = 10.5$ cm; TSA $= 2646 + 346.5 = 2992.5$ cm^2
5	**(c)** 15840 cm²	TSA = CSA cylinder + one flat base + CSA of cone; $= 9504 + 1386 + 4950 = 15840$ cm^2
6	**(a)** 2178 cm²	$2\pi rh + 2(2\pi r^2)$ with $r = 10.5$, $h = 12$; TSA $= 792 + 1386 = 2178$ cm^2
7	**(d)** 54	Volume of a sphere $= \frac{4}{3}\pi(3)^3$; $n = \frac{3r^2h}{4\rho^3}$; $n = 54$
8	**(b)** 6 cm	$\frac{4}{3}\pi R^3 = \frac{1}{3}\pi(4)^2(54)$ gives $R^3 = 216$; $R = 6$ cm
9	**(a)** 9592 cm²	TSA = CSA of cone + CSA of hemisphere $= \pi rl + 2\pi r^2$; TSA $= \frac{22}{7} \times 28 \times 53 + 2 \times \frac{22}{7} \times 28^2 = 9592$ cm^2
10	**(d)** 2178 cm²	$2\pi rh + 2(2\pi r^2) = 2 \times \frac{22}{7} \times 10.5 \times 12 + 4 \times \frac{22}{7} \times 10.5^2$; TSA $= 2178$ cm^2

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Volume of water displaced = volume of sphere = rise × base area; $\frac{4}{3}\pi(6)^3 = \pi(6)^2x$ (1); $x = 8$ cm (1).	1+1
12	Volume of sphere = volume of wire (cylinder) $= \pi r^2 L$; $\frac{4}{3}\pi(9)^3 = \pi(3)^2L$ (1); $L = 108$ cm (1).	1+1
13	Volume of cuboid $= n\times$ volume of a cube; $n = \frac{24\times24\times9}{3^3}$ (1); $n = 192$ (1).	1+1
14	Volume of water displaced = volume of sphere = rise × base area; $\frac{4}{3}\pi(9)^3 = \pi(9)^2x$ (1); $x = 12$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Volume = hemisphere + cone (1); Hemisphere $= \frac{2}{3}\pi r^3 = 19404$ cm^3; cone $= \frac{1}{3}\pi r^2 h = 16170$ cm^3 (1); Volume $= 35574$ cm^3 (1).	1+1+1
16	Volume = cylinder + two hemispheres $= \pi r^2 h + \frac{4}{3}\pi r^3$ (1); Cylinder $= 5197.5$ cm^3; two hemispheres $= 4851$ cm^3 (1); Volume $= 10048.5$ cm^3 (1).	1+1+1
17	Volume left = cylinder − cone (1); Cylinder $= 3696$ cm^3; cone $= 1232$ cm^3 (1); Volume left $= 2464$ cm^3 (1).	1+1+1
	OR: Volume = hemisphere + cone (1); Hemisphere $= \frac{2}{3}\pi r^3 = 2425.5$ cm^3; cone $= \frac{1}{3}\pi r^2 h = 2425.5$ cm^3 (1); Volume $= 4851$ cm^3 (1).	1+1+1
18	Volume = cylinder + two hemispheres $= \pi r^2 h + \frac{4}{3}\pi r^3$ (1); Cylinder $= 13860$ cm^3; two hemispheres $= 38808$ cm^3 (1); Volume $= 52668$ cm^3 (1).	1+1+1
	OR: Volume left = cylinder − cone (1); Cylinder $= 16632$ cm^3; cone $= 5544$ cm^3 (1); Volume left $= 11088$ cm^3 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Volume of water is the same in both tanks (2); $\pi r^2 h = 3696$ cm^3 and base area of cuboid $= 308$ cm^2 (2); Height $= 3696 \div 308 = 12$ cm (1).	2+2+1
	OR: Volume of water in the cone $= \frac{1}{3}\pi r^2 h$ (2); $\frac{1}{3}\pi(6)^2(24) = \pi(4)^2x$ (2); $x = 18$ cm (1).	2+2+1
20	Volume of water is the same in both tanks (2); $\pi r^2 h = 4312$ cm^3 and base area of cuboid $= 308$ cm^2 (2); Height $= 4312 \div 308 = 14$ cm (1).	2+2+1
	OR: Volume of water in the cone $= \frac{1}{3}\pi r^2 h$ (2); $\frac{1}{3}\pi(6)^2(30) = \pi(6)^2x$ (2); $x = 10$ cm (1).	2+2+1

Mistakes to avoid:

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- Using the vertical height instead of the *slant* height in the CSA and TSA of a cone.
- Forgetting the $\frac{1}{3}$ in the volume of a cone or using $2\pi r^2$ for the surface area of a full sphere.
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